

DistilBERT for Skincare Review Recommendation Prediction: A Comparative Study of Traditional Machine Learning and Transformer-Based Methods

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Abstract— This project evaluates transformer-based deep learning for skincare review recommendation prediction using the Sephora Products and Skincare Reviews dataset. TF-IDF Logistic Regression and Linear SVM baselines were compared with a fine-tuned DistilBERT model and a metadata-enhanced hybrid variant. The cleaned dataset contained 227,643 reviews and was split using stratified sampling. DistilBERT achieved the strongest performance with an F1-score of 0.969, outperforming both traditional

machine learning baselines. The hybrid model remained competitive but did not surpass the text-only transformer. These results demonstrate the effectiveness of contextual transformer representations for skincare review classification

Keywords— DistilBERT, skincare reviews, recommendation prediction, natural language processing, TF-IDF, transformer models.

I. INTRODUCTION

Online beauty retail platforms such as Sephora contain large volumes of customer reviews that strongly influence purchasing decisions. Consumers often rely on review text, ratings, and product-related context when selecting skincare products for concerns such as dryness, sensitivity, acne, and oiliness. However, manually analysing large volumes of reviews is time-consuming, which motivates the use of natural language processing (NLP) models for automated review understanding. Review-based recommender research has shown that user-generated text can complement or improve on rating-only approaches, but extracting useful signals from reviews remains challenging [1].

Recent advances in transformer-based NLP, particularly BERT and DistilBERT, have substantially improved contextual language understanding for text classification tasks [2], [3]. BERT introduced deeply bidirectional pre-training for language representation learning [2], while DistilBERT offers a lighter alternative that is approximately 40% smaller and 60% faster than BERT while retaining much of its performance [3]. In parallel, recent skincare and beauty recommendation studies have explored ingredient-aware recommendation and explainable beauty attributes, demonstrating that domain-specific signals are important for beauty applications [4], [5]. While previous research has investigated review-based recommendation, transformer-based NLP, and ingredient-aware skincare recommendation systems [1]–[5], many approaches focus on rating prediction, ingredient analysis, or explainability rather

than directly evaluating whether compact transformer models can improve binary recommendation prediction from customer review text. Furthermore, limited attention has been given to whether lightweight skincare-review metadata, such as skin type and review length, provides additional predictive value when combined with contextual transformer embeddings. This project addresses that gap by comparing TF-IDF baselines, text-only DistilBERT, and a metadata-enhanced hybrid model under a controlled experimental setting.

The aim of this project is to evaluate transformer-based deep learning for skincare review recommendation prediction and investigate whether lightweight skincare-review metadata improves model performance. The project compares TF-IDF baselines with transformer-based deep learning, where Logistic Regression and linear SVM are implemented as classical baselines and DistilBERT is fine-tuned as an encoder-only transformer model with a linear classification head for contextual review understanding. A hybrid variant is then evaluated by combining DistilBERT text representations with lightweight metadata, specifically skin type and derived review length. The input to the primary system is customer review text, with skin type and review length added in the hybrid setting, and the output is a binary label indicating whether a product is recommended or not recommended. Controlled experiments demonstrate that text-only DistilBERT achieves the strongest F1-score, outperforming the controlled TF-IDF baselines, while the metadata-enhanced hybrid remains competitive but does not surpass the text-only transformer. Model performance is evaluated using accuracy, precision, recall, and F1-score.

II. DATASET

The experiments were conducted using the publicly available *Sephora Products and Skincare Reviews* dataset obtained from Kaggle [6]. The original dataset contains approximately 8,494 beauty products and over 1 million customer reviews collected from Sephora’s online platform. Since this project focuses on skincare recommendation prediction, only products belonging to the *Skincare* category were retained. After merging the product and review datasets and applying preprocessing, the final cleaned dataset contained 2,224 skincare products and 227,643 customer reviews.

Each review entry included review text, recommendation labels, ratings, and lightweight metadata such as skin type, while product entries included information such as product name, brand, ingredients, and category. The binary target variable was *is_recommended*, where 1 represented recommended products and 0 represented not recommended products. A derived feature, *review_length*, was also created by calculating the number of words in each review.

Several preprocessing steps were performed to improve data quality and ensure reproducibility. Products with missing ingredient or review-related information were removed, while missing categorical metadata such as *skin_type* were replaced with the token *Unknown*. Duplicate reviews and reviews with missing text or missing recommendation labels were removed. Extremely short reviews containing fewer than three words were also discarded because they provided little semantic information for transformer-based language modelling. Text was intentionally left largely unprocessed because DistilBERT performs its own contextual tokenisation internally.

The final dataset remained moderately imbalanced, with approximately 84.4% recommended reviews and 15.6% not recommended reviews. To ensure fair evaluation, stratified splitting was applied using a fixed random seed of 42. The cleaned dataset was divided into 64% training, 16% validation, and 20% testing subsets, producing 145,691 training reviews, 36,423 validation reviews, and 45,529 test reviews. Stratification preserved the original class distribution across all splits and improved experimental reliability.

Table I. Dataset Statistics

Statistic	Value
Final skincare products	2,224
Final reviews	227,643
Recommended reviews	192,080 (84.4%)
Not recommended reviews	35,563 (15.6%)
Training samples	145,691
Validation samples	36,423
Testing samples	45,529

The dataset is suitable for transformer-based NLP because of its large review volume, real-world user-generated language, and availability of skincare-related contextual metadata. However, the dataset may contain demographic and platform-specific biases because reviews originate from self-selected Sephora customers. In addition, skin-type information is self-reported and therefore may not always be

fully reliable. Despite these limitations, the dataset provides a realistic benchmark for evaluating metadata-enhanced transformer models for skincare recommendation prediction

III. METHODOLOGY

A. Model Architecture

This project evaluates both traditional machine learning baselines and transformer-based deep learning models for skincare review recommendation prediction. TF-IDF with Logistic Regression and linear SVM were implemented as classical NLP baselines because they are computationally efficient and widely used for text-classification tasks [7], [8]. The primary deep learning model is DistilBERT [3], a compact transformer architecture derived from BERT [2].

DistilBERT is an encoder-only transformer model that processes tokenised review text through stacked self-attention encoder layers. Each encoder layer consists of multi-head self-attention and feed-forward sublayers combined with residual connections and layer normalisation. During tokenisation, a special [CLS] token is added to the beginning of each review sequence. The final hidden representation of this token is treated as a contextual representation of the entire review and is passed into a linear classification head for binary recommendation prediction.

A second hybrid architecture was also evaluated. In this variant, the contextual [CLS] embeddings extracted from DistilBERT were concatenated with lightweight skincare-review metadata, specifically review length and skin type. The combined feature representation was then classified using Logistic Regression. This hybrid design was introduced to investigate whether skincare-specific contextual metadata could improve recommendation prediction while remaining computationally lightweight.

B. Loss Function

The DistilBERT classifier was fine-tuned using the standard cross-entropy loss function for binary classification [3]. Given predicted probability $\hat{y}$ and true class label y , the loss is defined as:

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

Cross-entropy loss penalises incorrect predictions while rewarding higher confidence for correct classifications. Since the dataset is moderately imbalanced, F1-score was selected as the primary evaluation metric because it balances both precision and recall more effectively than accuracy alone.

C. Optimization Process

DistilBERT was fine-tuned using the AdamW optimiser with a learning rate of 2×10^{-5} , batch size of 16, and 3 training epochs. Weight decay of 0.01 was applied to reduce overfitting during fine-tuning. The HuggingFace TrainerAPI was used with validation evaluation performed after each epoch, while the checkpoint with the highest validation F1-score was automatically retained for final testing.

Figure 1 illustrates the validation F1-score achieved by DistilBERT across the three fine-tuning epochs. Performance

improved consistently from 0.967 in the first epoch to approximately 0.971 in the final epoch, indicating stable convergence during training. The relatively small performance gains observed after the second epoch suggest that the model was approaching convergence, supporting the decision to use three training epochs and retain the checkpoint with the highest validation F1-score for final evaluation.

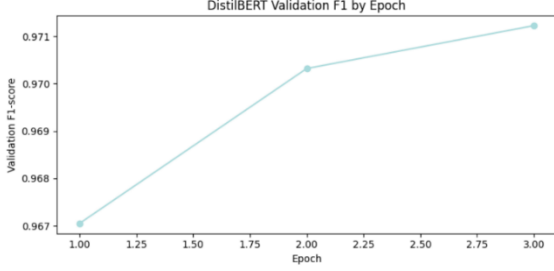


Fig. 1. DistilBERT validation F1-score during fine-tuning

For the TF-IDF baselines, text features were generated using unigram and bigram TF-IDF representations with a maximum vocabulary size of 10,000 features. Logistic Regression and linear SVM were trained using balanced class weighting to reduce the effect of class imbalance.

In the hybrid model, DistilBERT embeddings were first extracted and frozen before being concatenated with metadata features. Logistic Regression with the liblinear solver was then trained on the combined feature representation

D. Mathematical Representation

Given tokenised review input X , DistilBERT produces contextual hidden representations:

$$H = \text{DistilBERT}(X)$$

The contextual representation of the [CLS] token, denoted as h_{CLS} , is extracted from the final hidden layer and passed into a linear classifier:

$$z = Wh_{CLS} + b$$

Prediction probabilities are calculated using the classifier output logits and converted into class probabilities using the softmax function.

$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

For the hybrid architecture, metadata vector m containing review length and skin-type information is concatenated with the DistilBERT embedding:

$$h_{\text{hybrid}} = [h_{CLS}; m]$$

The combined representation is then used for final classification

E. Model Workflow

Figure 2 illustrates the overall workflow of the proposed metadata-enhanced DistilBERT framework. Review text is first processed through the DistilBERT tokenizer and encoder before classification. For the hybrid model, review length and skin-type metadata are concatenated with

the DistilBERT [CLS] representation prior to final prediction.

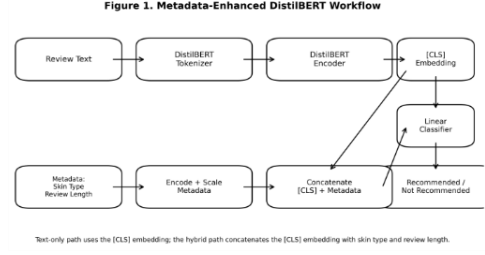


Fig. 2. Metadata-Enhanced DistilBERT Workflow

F. Training Procedure

DistilBERT was fine-tuned using the AdamW optimiser with a learning rate of 2×10^{-5} , batch size of 16, and three training epochs. Model performance was monitored on the validation set after each epoch, and the checkpoint achieving the highest validation F1-score was retained for final testing. For the TF-IDF baselines, unigram and bigram features were generated using a vocabulary size of 10,000 features. Logistic Regression and Linear SVM were trained using balanced class weighting to mitigate the effects of class imbalance. For the hybrid approach, DistilBERT embeddings were extracted and concatenated with review-length and skin-type metadata before classification using Logistic Regression.

IV. EXPERIMENT

A. Evaluation Metrics

Model performance was evaluated using Accuracy, Precision, Recall and F1-score. Because the dataset was moderately imbalanced, with 84.4% of reviews belonging to the Recommended class, F1-score was selected as the primary evaluation metric. Unlike accuracy, F1-score balances both precision and recall, providing a more reliable assessment of classification performance on imbalanced datasets.

B. Quantitative Results

Table II summarises the performance of all evaluated models on the controlled test subset. The TF-IDF baselines achieved strong results, with Logistic Regression obtaining an F1-score of 0.931 and Linear SVM achieving an F1-score of 0.941. These findings demonstrate that traditional machine learning models remain effective for large-scale review classification when combined with high-dimensional text representations.

TABLE II

Performance of All Models on the Controlled Test Subset

Model	Acc.	Prec.	Recall	F1
LR	0.888	0.964	0.900	0.931
SVM	0.902	0.953	0.930	0.941
DistilBERT	0.947	0.969	0.969	0.969
Meta	0.651	0.842	0.723	0.778

Hybrid	0.941	0.971	0.959	0.965
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LR = Logistic Regression, SVM = Linear Support Vector Machine, Hybrid = DistilBERT + metadata.

The fine-tuned DistilBERT model achieved the highest overall performance, obtaining an accuracy of 94.7% and an F1-score of 0.969. Compared with the strongest baseline model, DistilBERT improved F1-score by approximately 2.8 percentage points. Although the numerical improvement appears relatively small, it is meaningful given the already strong performance of the TF-IDF baselines and the large scale of the dataset. The results indicate that contextual transformer representations provide additional predictive capability beyond traditional sparse text features. This improvement suggests that contextual transformer representations provide richer semantic understanding than sparse TF-IDF features, enabling the model to capture relationships between words that are difficult for traditional approaches to represent.

Figure 2 illustrates the performance comparison across all evaluated models. DistilBERT consistently outperformed both classical machine learning baselines, confirming the effectiveness of transformer-based contextual language modelling for skincare review recommendation prediction.

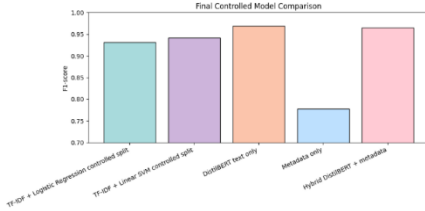


Fig. 3. F1-score comparison chart

C. Analysis of the Metadata-Enhanced Hybrid Model

To investigate whether lightweight skincare-review metadata could improve transformer performance, a hybrid architecture was evaluated using DistilBERT embeddings combined with review length and skin-type metadata. The hybrid model achieved an F1-score of 0.965, which was slightly lower than the 0.969 obtained by the text-only DistilBERT model. This suggests that the additional metadata did not provide enough independent signal to improve performance beyond the contextual information already learned from review text.

The metadata-only model performed substantially worse than all text-based approaches, further supporting the conclusion that customer review text was the dominant predictive feature in this dataset. Consequently, the proposed hybrid architecture did not outperform the simpler text-only transformer and was therefore not selected as the final model.

D. Error Analysis and Discussion

The confusion matrix for the final DistilBERT model is shown in Figure 4. The model correctly classified 390 Not Recommended reviews and 2452 Recommended reviews, while only 158 reviews were misclassified across both classes. This demonstrates strong generalisation performance despite the moderate class imbalance present in the dataset. Although DistilBERT achieved the strongest overall performance, the confusion matrix shows that some Not Recommended reviews were still predicted as Recommended. This behaviour is expected given the class imbalance of the dataset and the presence of reviews containing mixed positive and negative sentiment, which can make classification more challenging.

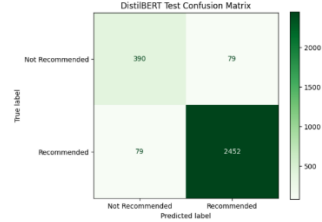


Fig. 4. DistilBERT Test Confusion Matrix.

A detailed inspection of misclassified examples revealed that many errors occurred in reviews containing mixed sentiment or contradictory language. For example, reviews describing positive product performance while simultaneously reporting minor side effects were occasionally misclassified. Such cases are inherently difficult because the recommendation label does not always align perfectly with the sentiment expressed within the review text.

Overall, the experimental results demonstrate that transformer-based contextual modelling provides a significant advantage over traditional TF-IDF approaches for skincare review recommendation prediction. DistilBERT achieved the strongest performance across all evaluation metrics and therefore represents the most effective model evaluated in this study.

V. CONCLUSION

This study evaluated transformer-based deep learning for skincare review recommendation prediction using the Sephora Products and Skincare Reviews dataset. DistilBERT achieved the strongest performance, obtaining an F1-score of 0.969 and outperforming both TF-IDF Logistic Regression and Linear SVM baselines. These results demonstrate the effectiveness of contextual transformer representations for understanding customer review text.

The metadata-enhanced hybrid model did not surpass the text-only DistilBERT model, suggesting that review text already contained most of the predictive information required for recommendation prediction. A limitation of this study is that only a single dataset was evaluated and transformer experiments were conducted on a controlled subset due to computational constraints. Future work could investigate larger transformer architectures, multimodal product information, and additional beauty-review datasets.

The source code, experimental notebook, and supporting documentation were managed using GitHub to support reproducibility and open-source accessibility.

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